Experiment 9

Logistic Regression.

Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df
df.head()

features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

for i in range(1,401):
    x_train,x_test,y_train,y_test=train_test_split(
        features,label,test_size=0.2,random_state=i)
    model=LogisticRegression()
    model.fit(x_train,y_train)
    train_score=model.score(x_train,y_train)
    test_score=model.score(x_test,y_test)
    if test_score>train_score:
        print("Test {} Train{} Random State {}".format(
            test_score,train_score,i))

x_train,x_test,y_train,y_test=train_test_split(
    features,label,test_size=0.2,random_state=37)

finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)

print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))

from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

Output

```
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0

400 rows × 5 columns
```

```
   User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510  Male     19  19000             0
1  15810944  Male     35  20000             0
2  15668575  Female   26  43000             0
3  15603246  Female   27  57000             0
4  15804002  Male     19  76000             0
```

Output

```
Test 0.6875 Train0.63125 Random State 3
Test 0.7375 Train0.61875 Random State 4
Test 0.6625 Train0.6375 Random State 5
Test 0.65 Train0.640625 Random State 6
Test 0.675 Train0.634375 Random State 7
Test 0.675 Train0.634375 Random State 8
Test 0.65 Train0.640625 Random State 10
Test 0.6625 Train0.6375 Random State 11
Test 0.7125 Train0.625 Random State 13
Test 0.675 Train0.634375 Random State 16
Test 0.7 Train0.628125 Random State 17
Test 0.7 Train0.628125 Random State 21
Test 0.65 Train0.640625 Random State 24
Test 0.6625 Train0.6375 Random State 25
Test 0.75 Train0.615625 Random State 26
Test 0.675 Train0.634375 Random State 27
Test 0.7 Train0.628125 Random State 28
Test 0.6875 Train0.63125 Random State 29
Test 0.6875 Train0.63125 Random State 31
Test 0.6625 Train0.6375 Random State 37

LogisticRegression()

0.865625

0.8125

              precision    recall  f1-score   support

           0       0.85      0.93      0.89       257
           1       0.84      0.71      0.77       143

    accuracy                           0.85       400
   macro avg       0.85      0.82      0.83       400
weighted avg       0.85      0.85      0.85       400
```